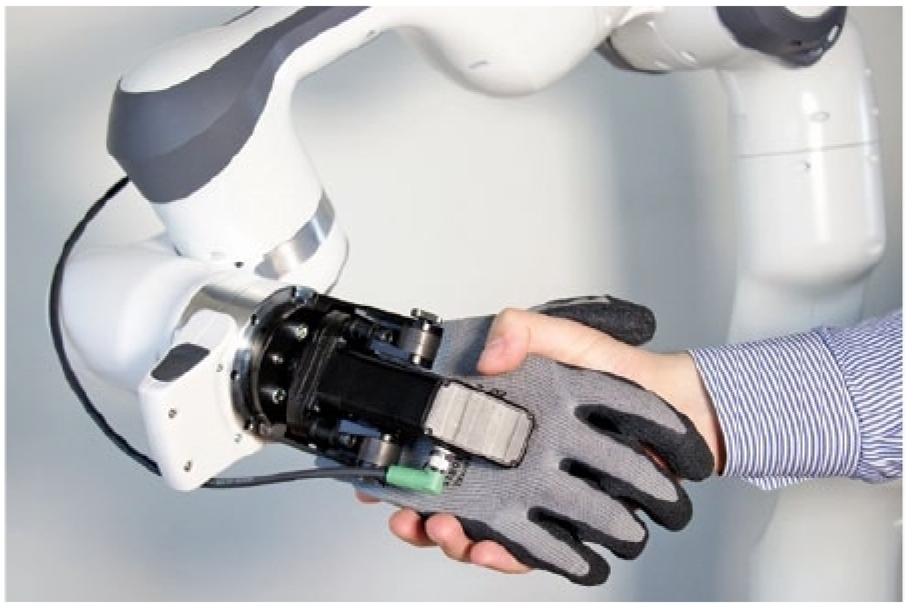
biomedical engineering at University of Virginia, is one such researcher. His work combines computer science and biology to understand viral infections such as influenza and Zika.

His research team observes viruses experimentally by manually tagging them with fluorescent proteins and viewing them through microscopes to understand how they affect cells. However, this is a difficult and time-consuming process with room for human error. Enter supercomputer Frontera, the most powerful academic supercomputer in the world.

In addition to studying viruses experimentally, the supercomputer allows Kasson's team to take apart critical viral mechanisms and understand them at the atomic level using simulations—virtual re-creations of a virus's behaviour.

From these new insights, the scientists want to track and combat viruses. "What we do guides the development of new antiviral therapies and also helps us assess how well vaccines work and how well people's immunity can prevent new viral threats from causing widespread disease in the US," he added.

Anatomically-shaped hand for the intelligent tool Panda



With nineteen dislocatable, self-healing finger joints, SoftHand replaces numerous individual grippers and, thus, is predestined for complex handling tasks (Credit: German Robotics GmbH)

With their SoftHand, German Robotics from Munich present the first anatomically-shaped hand for collaborating robots (cobots). SoftHand for research is intended for educational purposes, research institutes, laboratories and universities. It is available in different colours and is equipped with a stiff wrist. Numerous software interfaces, including Matlab-Simulink and a high-level C interface, are available to simplify integration with test programs.

The industrial-grade variant impresses with its flexible application options, which avoid tiresome gripper changes—between the thumb and a finger, SoftHand reliably holds objects weighing up to 600 grams. Once SoftHand is fully utilised, maximum gripping weight increases to two kilograms. The wrist is rotatable, and gloves are easily replaceable. Protection class IP65 makes it possible to use it in damp, dusty or other environments that are unpleasant for humans. SoftHand fulfils numerous normative requirements, especially regarding risk assessment, strength and force limitations.

Especially for highly-sensitive robots, such as Panda by Franka Emika, which has force sensors in all seven joints, SoftHand is a congenial supplement. With its anatomical shape and nineteen finger joints, SoftHand can be used as a gripper for parts with complex geometries as well as for shaking electromechanical or large electronic components into place in electronics production.

CloudMinds XR-1, one of the first intelligent 5G humanoid robots comes to life

CloudMinds Technology Inc., a global pioneer in cloud artificial intelligence (AI) architecture that makes robots and businesses smarter, had its revolutionary XR-1 robot interact with guests at Sprint exhibit at Mobile World Congress Los Angeles, held from October 22 to 24. XR-1 is one of the first-ever humanoid robots powered by cloud AI, commercial Sprint True Mobile 5G and proprietary vision-controlled grasping technology for service robots that also leverages human operator input for constant learning.

"Overall, intelligent cloud robots paint the most vibrant picture of how 5G's ultra-low latency, exponentially faster speeds and wider reach can dramatically improve response time and enable a new world of applications," said Bill Huang, founder and chief executive officer, CloudMinds. "With vision-controlled grasping and the ability to perform intricate tasks, XR-1 simply raises the bar and lays the foundation for an even wider range of intelligent compliant cloud service robots from CloudMinds—from wheeled to two-legged form factors. We are proud to be ushering in a new era of helpful robots for homes and businesses, with an emphasis on the importance of human input."

XR-1 is capable of answering questions (with voice and emotion recognition), serving drinks, threading needles and dancing (Credit: https://finance.yahoo.com)